

Short Paper

An Empirical Study Challenging the Ability of Heterogeneous Graph Neural Networks to Effectively Learn Topological Structures

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Abstract—The utilization of graph neural networks in heterogeneous networks has been rapidly increasing in recent years. The topological structure, fundamental to the nature of graphs, is often claimed to be jointly addressed with attribute information by most heterogeneous graph neural network (HGNN) approaches. However, this assertion lacks sufficient empirical validation. We empirically challenge these claims within existing evaluation frameworks. We propose a straightforward multi-layer perceptron (MLP) based model which relies solely on adjacency relationships for feature aggregation, without fully exploiting the graph's intricate topological structures. Unexpectedly, the model achieved nearly state-of-the-art performance on several real-world datasets. We further validate our findings through topology-sensitive synthetic experiments on Hetero-House and higher-order relation prediction tasks, confirming that current evaluation protocols contain systemic biases and often fail to distinguish superficial adjacency exploitation from genuine structural pattern learning. This points to the need for future benchmark datasets and metrics that more faithfully reflect the real requirements of topological learning. We hope this work stimulates further research into the role and significance of topological information in HGNNs.

Index Terms—Heterogeneous graph, graph neural networks, graph topology, evaluation protocols.

I. INTRODUCTION

Graphs serve as versatile data structures for representing entities and their complex interrelations, finding extensive application in fields ranging from social media analysis [1] to industrial applications [2]. Crucially, edges within these graphs interconnect to form intricate topological structures, such as cycles and cliques [3], [4], constituting the graph's essence and are vital in practical scenarios.

To jointly handle attribute and topological information, graph neural network (GNN) architectures have been widely introduced [5]. Their message-passing mechanism embeds graph attributes and topology

into the representation space. However, the influence of attributes and topology on downstream tasks is often imbalanced. Attribute information dominates in recommendation systems [6] while topological information proves more crucial in community detection [7].

Researchers have enhanced GNNs to better capture graph structure and highlight topological importance. Some approaches apply complex Weisfeiler-Leman tests to differentiate graph architectures [8]. Others incorporate subgraph isomorphism counting to represent local structure details [9] or pre-process local neighborhoods [10]. Additionally, minor structure perturbations assist the network in recognizing critical variations [11]. These methodologies lay a solid foundation for the implementation of GNNs in elaborate graph scenarios and facilitate the advancement of more topology-aware GNNs.

Real-world graphs are typically composed of multiple types of nodes and edges [12]. For example, social networks contain users, posts, and comments with relationships encompassing various categories like commenting or following. Classical GNN methods assume homogeneous nodes and edges, limiting their ability to handle complex heterogeneous interactions [13]. Consequently, heterogeneous graph neural networks (HGNNs) integrate type information through specialized aggregation strategies to effectively model heterogeneous relationships.

Most HGNN methods routinely integrate both attribute and topological information. As illustrated in Fig. 1, attribute-driven learning focuses on the propagation and aggregation of neighborhood attribute information. Topological representation learning focuses on how to identify the topological structure and treats the graph structure as an optimizable learning target. However, a critical distinction is often overlooked. Most HGNNs claim to model graph topology, but primarily use graph structure only for message passing. These attribute-driven models aggregate neighbor features via adjacency without capturing intrinsic structural patterns like motifs or cycles. This conflation creates a significant gap in our understanding. Do current downstream tasks actually require complex topological reasoning? More critically, can existing benchmarks effectively distinguish between superficial adjacency exploitation and genuine structural pattern learning?

To empirically expose this gap, we introduce the Minimal Topological Heterogeneous Graph Network (MTHGN) as a controlled baseline. This model operates on the principle of minimal topology, characterized by the utilization of direct adjacency relationships strictly as a conduit for feature aggregation. In this paradigm, the graph structure functions merely as a static channel for message passing, devoid of higher-order semantic encoding. This design sharply contrasts with architectures that leverage complex topology, such as those explicitly extracting meta-paths, identifying network motifs, or learning dynamic structures. By restricting MTHGN to these minimal signals, we isolate the impact of superficial connectivity from that of genuine structural learning.

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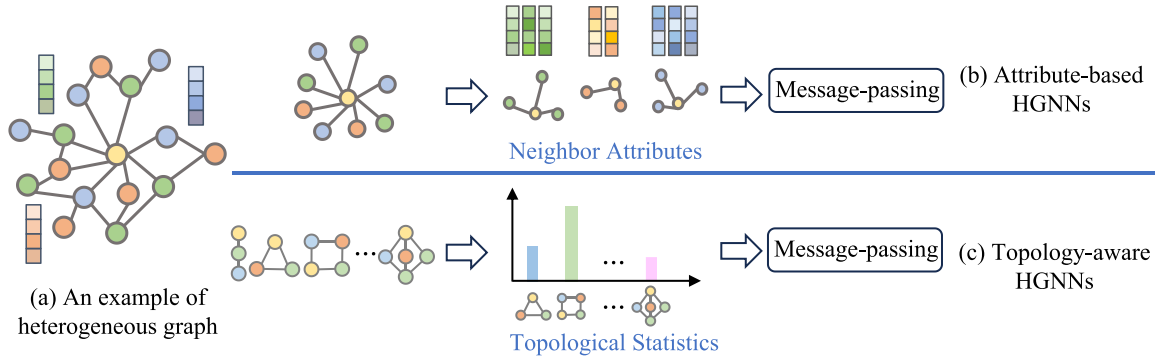


Fig. 1. Attribute-based HGNNs learn node representations through type-specific message passing and aggregation, where the updates mainly rely on the attribute information of the neighborhood. Topology-aware HGNNs extract type-aware high-order structural signals such as motifs, graphlets, and role or spectral features to guide the representation learning process.

Despite this simplification, MTHGN achieves near state-of-the-art (SOTA) results on real-world datasets, suggesting that popular heterogeneous benchmarks rely predominantly on feature and neighbor information under current evaluation protocols. We further validated this through multiple perspectives: (1) we investigated the impact of expanding the receptive field (generally assumed to capture more structural information) by integrating ordinary differential equations (ODEs) [14] to mitigate over-smoothing in deeper networks, but found these enhanced models did not significantly surpass their shallower counterparts; (2) we conducted experiments on Hetero-House, a topology-sensitive synthetic dataset; (3) we evaluated higher-order relation prediction tasks, which revealed inconsistent patterns and exposed gaps between claimed and actual topological capabilities. These experiments confirm that current protocols fail to reward genuine topological learning.

The main contributions of this paper are:

- We uncover deficiencies in current HGNN evaluation protocols for validating the capability to capture topological structure features.
- We propose MTHGN as a controlled baseline that relies solely on attributes and minimal topological signals encoded through direct adjacency aggregation. Despite this simplicity, it achieves near-SOTA performance on multiple real-world benchmarks, empirically exposing that existing datasets and evaluation protocols fail to require complex topological reasoning.
- We design Hetero-House, a topology-sensitive synthetic dataset, and introduce higher-order relation prediction tasks to comprehensively validate our findings. Through comparative analysis, we reveal systematic biases in current evaluation frameworks and provide new directions for future HGNN research on topological learning.

II. BACKGROUND AND RELATED WORK

A. Heterogeneous Graph

We represent a heterogeneous graph as $G = \{\mathcal{V}, \mathcal{E}, \mathcal{T}^v, \mathcal{T}^e\}$, where $\mathcal{V}$ denotes the set of nodes, $\mathcal{E}$ represents the set of edges, $\mathcal{T}^v$ and $\mathcal{T}^e$ correspond to the sets of vertex types and edge types, respectively. Nodes are mapped to their types through the function $\phi: \mathcal{V} \rightarrow \mathcal{T}^v$. Relations are classified by the function ψ . When only considering local adjacencies, $\psi: \mathcal{E} \rightarrow \mathcal{T}^e$.

B. End-to-End Heterogeneous Graph Neural Network

End-to-end heterogeneous graph neural networks perform message passing and parameter learning simultaneously during training. Based

on how they capture heterogeneous information, these methods can be categorized into relation-specific methods, metapath-based methods, and topology-aware methods.

Relation-Specific Methods: These methods directly leverage the multiple relation types in heterogeneous graphs by designing distinct message passing mechanisms for each relation. Representative methods in this category include HGT [15], which introduces a novel heterogeneous mutual attention mechanism based on Transformers. SimpleHGN [16] employs a multi-layer GAT [17] network as its core architecture and generates attention values by combining node features with learnable edge-type embeddings. HeTa [18] designs a relation-specific attack framework and perturbs injected nodes and edges per relation to reveal relation-level weaknesses. Our proposed MTHGN also falls into this category, aggregating neighbor features to capture relation-specific information.

Metapath-Based Methods: Metapath-based approaches capture high-order semantic relationships by utilizing predefined meaningful paths connecting different nodes. For instance, MAGNN [19] leverages information from all nodes within a metapath instance, not just the endpoints. MECCH [20] constructs the adjacency structure for each node through metapath contexts and further integrates the information within these structures to update the representations of the nodes. In addition, PM-HGNN [21] exploits predefined meta-paths to build multi-scale semantic graphs and aggregates path-guided structural signals to enhance heterogeneous representation learning.

Topology-Aware Methods: While relation-specific and metapath-based methods primarily focus on how to process neighborhood attribute information through different relation types or predefined semantic paths, topology-aware methods concentrate on how to handle the neighborhood network topology itself. This class of heterogeneous graph learning approaches goes beyond predefined relations and metapaths, because manually designed metapaths rely heavily on domain knowledge and have limited expressive power. Therefore, topology-aware approaches capture structural information by automatically learning the graph structure or using more general topological patterns such as random walks and subgraphs. This makes the graph structure itself an optimizable learning target rather than a fixed topology.

HGSL [22] is a representative framework. It models the interaction between node features and the initial topology through a feature propagation graph while using a semantic graph to capture high-order meta path semantics, dynamically reconstructing the topology of a heterogeneous graph. HGK-GNN [23] provides another topology-aware design by incorporating heterogeneous graph kernels to capture the graph's topological properties. Meanwhile, TACO [24] introduces a

TABLE I
COMPARISON OF END-TO-END HGNNs AND PRE-COMPUTATION MODELS

	Model Architecture	Key Modules	Computational Complexity
End-to-end HGNN	End-to-end structure: • Multi-layer, Iterative	Train: Relations/meta-paths based-message passing	$O(\mathcal{E} D + \mathcal{V} RD^2)$
Pre-Computation Models	Phased structure: • One-time pre-computation • Tensor mapping	Pre-process: Fixed neighbor aggregation Encoder: Learnable encoder	$O(\mathcal{V} RD^2)$

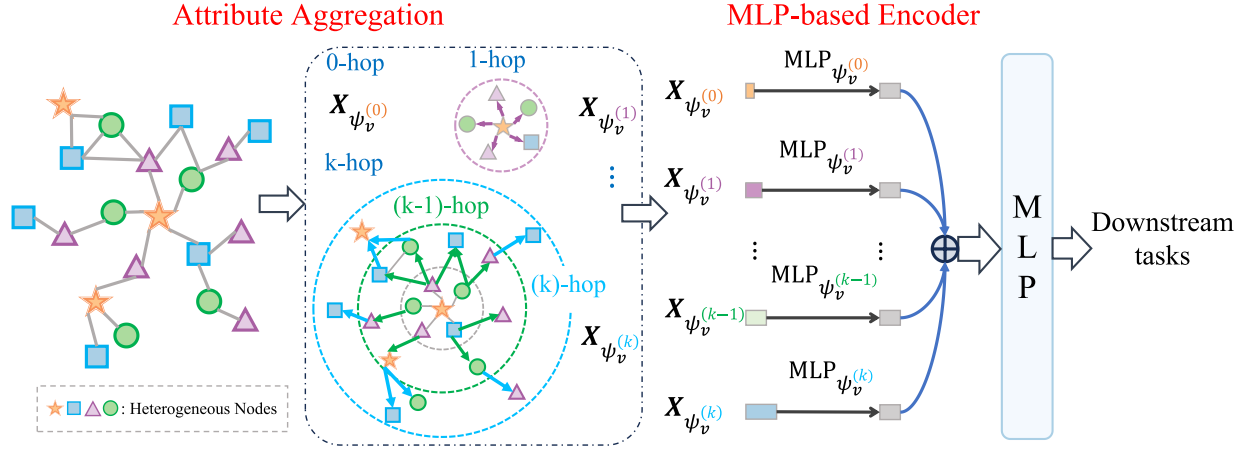


Fig. 2. MTHGN consists of two components: attribute aggregation and an MLP-based encoder. The model initially aggregates neighbor information, generating multi-hop feature representations. These features are then processed through feature projection and further integrated by multiple MLP layers to accommodate various downstream task requirements.

topology-aware graph framework that captures higher-order structural dependencies by learning subgraph-level patterns rather than relying on local neighbor aggregation.

C. Pre-Computation-Based Graph Learning Model

Table I lists the differences between the end-to-end HGNNs and the pre-computation models, where D represents the feature dimension, and R denotes the number of relation types. End-to-end HGNNs simultaneously perform parameterized message passing and feature learning. While pre-computation-based graph learning models adopt a phased structure, which first performs parameter-free information aggregation to enrich node features, then applies parameterized feature transformations, thereby reducing the computational complexity when message passing. Notably, our MTHGN belongs to the latter class. Given the advantages of pre-computation strategies, numerous studies have adopted this approach. SeHGNN [25] pre-computes mean features of neighboring nodes within different metapath scopes using a lightweight aggregator, then employs feature projections and a Transformer-based fusion module to integrate metapath-specific representations during training. RphGNN [26] pre-aggregates neighbor information via random projection iterations, then uses a learnable encoder for final representations.

III. METHOD

A. Overall Framework

Unlike existing pre-computation methods that employ sophisticated aggregation strategies or complex fusion modules, MTHGN adopts

a deliberately minimalist design in both pre-computation and training phases, as depicted in Fig. 2. Initially, the attribute aggregation phase utilizes a layered expansion of neighbor information method to aggregate neighbor information from diverse node types, yielding multi-dimensional node characteristics. A hierarchical-level projection MLP module then synthesizes these features more thoroughly. The enhanced feature vectors are subsequently concatenated and processed through an MLP layer tailored for final adaptation to diverse downstream requirements.

B. Attribute Aggregation

In the MTHGN model, we adopt a streamlined attribute aggregation strategy that significantly reduces computational overhead. MTHGN uses a hierarchical propagation strategy that recursively aggregates the features of k -hop neighbors based solely on adjacency information. This approach bypasses artificial semantic path construction and directly leverages the graph's natural connectivity structure for efficient feature propagation.

For $v \in V$ and its $(l-1)$ -hop neighbors $k \in \mathcal{N}_v^{(l-1)}$, the simplified neighbor aggregation process can be expressed as:

$$\mathbf{X}_{\psi_v^{(l)}} = \mathbf{D}_{\phi(k)}^{-1} \mathbf{A}_{\phi(k)} \mathbf{X}_{\psi_v^{(l-1)}}. \quad (1)$$

Here, $\psi_v^{(l)} = \psi(v, v_1, \dots, v_l) = \langle \phi(v), \dots, \phi(v_i), \dots, \phi(v_l) \rangle$. $(v, v_1, \dots, v_l)$ represents the sequence of nodes that node v passes through in the (l) -hop transition process. $\mathbf{X}_{\psi_v^{(l)}}$ is the (l) -th order semantic matrix obtained after l steps of transitions starting from node

Algorithm 1: Training Process of MTHGN.

Require: Raw feature matrices $\{X^{c_i} : c_i \in \mathcal{T}^v\}$; adjacency matrices $\{A_{\phi(k)}\}$; raw label matrix Y ; maximum hop number L

1: **// Phase 1: Pre-computation**

2: **for** each node type $c_i \in \mathcal{T}^v$ **do**

3: $X_{\psi_v^{(0)}} \leftarrow X^{c_i}$

4: **for** $l = 1$ to L **do**

5: $X_{\psi_v^{(l)}} \leftarrow D_{\phi(k)}^{-1} A_{\phi(k)} X_{\psi_v^{(l-1)}}$ for all connected neighbor types

6: **end for**

7: **end for**

8: **// Phase 2: MLP-based Training**

9: **for** each epoch **do**

10: $H^{\psi^{(l)}} \leftarrow \text{MLP}_{\psi^{(l)}}(X_{\psi^{(l)}})$ for $l = 0$ to L

11: $H^c \leftarrow [H^{\psi^{(0)}} || H^{\psi^{(1)}} || \dots || H^{\psi^{(L)}}]$

12: $\text{Pred} \leftarrow \text{MLP}_{\text{task}}(H^c)$; $\mathcal{L} = -\sum_i y_i \ln \text{pred}_i$

13: Back-propagate and update MLP parameters

14: **end for**

15: **return** Pred

v . $A_{\phi(k)}$ is the adjacency matrix from the $(l-1)$ -hop neighbors to the (l) -hop neighbors. $D_{\phi(k)}^{-1}$ is the inverse of the degree matrix of $A_{\phi(k)}$.

Starting from the original features of the initial node v , by consecutively applying the iteration formula, semantic representations at depth l can be obtained:

$$X_{\psi_v^{(l)}} = \left(\prod_{i=0}^l D_{\phi(k)}^{-1} A_{\phi(k)} \right) X_{\psi_v^{(0)}}, k \in \mathcal{N}_v^{(i)}. \quad (2)$$

Here, $\mathcal{N}_v^{(i)}$ represents the (i) -hop neighbor nodes of node v . When $l = 0$, $X_{\psi_v^{(0)}}$ directly reflects the raw feature matrix of node v , denoted as X_v . For the first-order aggregation, each node aggregates the attributes of its one-hop neighbors. Its main objective is to collect the local structural information directly associated with the node. This first-order aggregation enables the model to quickly grasp the node's direct neighborhood. For the second-order aggregation, the model handles second-order neighbors linked via first-order neighbors. This extended aggregation leverages various semantic relationships to enrich the neighborhood information being processed.

C. MLP-Based Encoder

During each training epoch, subsequent to the pre-computing phase, the hierarchically aggregated features of the nodes are projected via an MLP, transforming these features into new representations that are better suited for downstream tasks. Specifically, for the aggregated features of each level, the projection process can be described as:

$$h_{\psi_v^{(i)}} = \text{MLP}_{\psi_v^{(i)}}(X_{\psi_v^{(i)}}). \quad (3)$$

Here, $\text{MLP}_{\psi_v^{(i)}}$ is a specific MLP designed for the aggregation at level i . It is composed of normalization, nonlinear activation, and linear layers.

This process is conducted independently for each set of aggregated information, generating a series of projected feature vectors $h_{\psi_v^{(i)}}$. Finally, these projected vectors are concatenated to form the final representation of the nodes:

$$h_v = \text{CONCAT}(\{h_{\psi_v^{(i)}} : i = 0, 1, \dots, l\}). \quad (4)$$

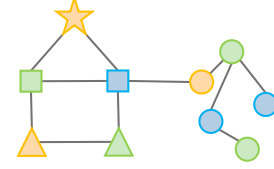


Fig. 3. A House motif from our Hetero-House dataset connected to a random Barabási-Albert Graph [27]. The colors indicate the node type (red for type A, blue for type B, green for type C). The shapes of House motif nodes indicate their topological roles (star for roof, square for wall, triangle for bottom). BA nodes are represented as circles. BA nodes do not participate in classification.

D. Optimization

In downstream tasks, a specific MLP is further used to transform the node representation into prediction results:

$$\hat{Y} = \text{MLP}_{\text{task}}(H). \quad (5)$$

Here, H is the representation matrix of all nodes, $\hat{Y}$ denotes the model's predicted output.

To optimize the model, a cross-entropy loss is defined during training to measure the discrepancy between the predicted results and the ground truth:

$$\mathcal{L} = -\frac{1}{|\mathcal{V}_T|} \sum_{v \in \mathcal{V}_T} \sum_{c=1}^C y_{vc} \log(\hat{y}_{vc}). \quad (6)$$

Here, $\mathcal{V}_T$ are training samples, and C is the number of classes. y_{vc} represents the ground truth label for node v , and $\hat{y}_{vc}$ denotes the probability predicted by the model.

IV. EXPERIMENTS

A. Datasets

Experiments are conducted on 4 heterogeneous graph datasets, including 3 real-world datasets and 1 synthetic dataset. The real-world datasets (DBLP, ACM, IMDB) are sourced from the HGB benchmark.¹ To evaluate the learning capability of HGNNs when topological structure is crucial for classification, we construct a topology-sensitive synthetic dataset called Hetero-House, as shown in Fig. 3.

Unlike existing benchmark datasets where node attributes are strongly correlated with labels, Hetero-House ensures that classification relies entirely on topological structure through the following mechanisms: 1) node features are Gaussian random vectors devoid of semantic information; 2) labels are determined solely by topological roles within House motifs; 3) node types are randomly assigned independently of topological roles, eliminating type information as a direct label indicator. The statistics of these datasets are summarized in Table II.

B. Baselines

We compare MTHGN with both end-to-end HGNN baselines including HAN [28], RGCN [29], HGT [15], SimpleHGN [16], HGSL [22] as well as pre-computation based model, SeHGNN. HAN captures hierarchical attention at node and semantic levels. RGCN applies relation-specific aggregation. HGT uses Transformer-based mutual attention. SimpleHGN enhances GAT with edge-type embeddings and residual connections. HGSL refines graph topologies by integrating feature propagation and semantic graphs. SeHGNN simplifies training via pre-computed neighbor aggregation and transformer-based semantic fusion.

¹<https://github.com/THUDM/HGB>

TABLE II
HETEROGENEOUS DATASET STATISTICS

Dataset	#Nodes	#Edges
DBLP	author (A):4,057, venue (V):20 term (T):7,723, paper (P):14,328	P-A:19,645 P-V:14,328 P-T:85,810
ACM	paper (P):3,025, term (T):1,902 subject (S):56, author (A):5,959	P-P:5,343 P-A:9,949 P-S:3,025 P-T:255,619
IMDB	movie (M):4,932, actor (A):6,124 director (D):2,393, keyword (K):7,971	M-D:4,932 M-A:14,779 M-K:23,610
Hetero- House	A:1,446, B:1,453 C:1,401	A-B:1,492 B-C:1,423 A-C:1,495

C. Implementation Details

We use multiple random seeds and run each method 10 times on all datasets. The results are reported as the mean scores along with their standard deviations. Key hyperparameters are configured as follows: the hidden layer dimension is set to 512, the learning rate is fixed at 0.001, and the dropout rate is 0.5. All benchmark methods are optimized using the Adam optimizer.

D. Experimental Results

For the node classification task, we adopt a transductive setting, with node labels split into training (24%), validation (6%), and testing (70%) sets. Evaluation is conducted using Macro-F1 and Micro-F1 scores for both multi-class (DBLP, ACM) and multi-label (IMDB, Hetero-House) datasets. For single-label classification, we employ a softmax function with cross-entropy loss. For the multi-label IMDB dataset, a sigmoid activation with binary cross-entropy loss is utilized. To comprehensively evaluate models' ability to capture complex topological structures, we introduce higher-order relation prediction [30]. This task aims to simultaneously predict the complete set of relationships among a given group of nodes. For this task, we supplement Macro-F1 and Micro-F1 with Normalized Mutual Information (NMI), a topology-sensitive metric, using cross-entropy loss as the objective function. Experimental results are presented in Tables III and IV, where the best-performing results are highlighted in bold and the second-best results are underlined.

On real-world datasets, SeHGNN and MTHGN achieve the best and second-best performance without utilizing GNN modules for topological learning, yet outperform end-to-end HGNNs. To mitigate interference from the powerful Transformer module used during SeHGNN's training, we use MTHGN as a control group for comparison. Notably, MTHGN relies solely on attribute features and minimal adjacency relations, yet achieves near-SOTA performance, strongly suggesting that on these attribute-rich real-world datasets, simple attribute matching and shallow neighbor aggregation may be the primary sources of performance.

Table IV presents higher-order relation prediction performance on real-world datasets. MTHGN and SeHGNN remain the top-performing models, while the topology-aware method HGSL also demonstrates excellent performance. As shown, HGSL demonstrates relatively poor performance on node classification, yet achieves excellent results on higher-order relation prediction. This discrepancy may be attributed to HGSL's emphasis on topological learning at the expense of effective attribute utilization. Conversely, attribute-driven baseline models perform poorly on higher-order relation prediction, revealing that they may not capture topological information as effectively as claimed.

Furthermore, Table III reveals that all models experience substantial performance degradation on the Hetero-House dataset. Relation-aware models such as HGT, RGCN, and SimpleHGN learn separate parameters for different relation types but perform poorly, indicating that

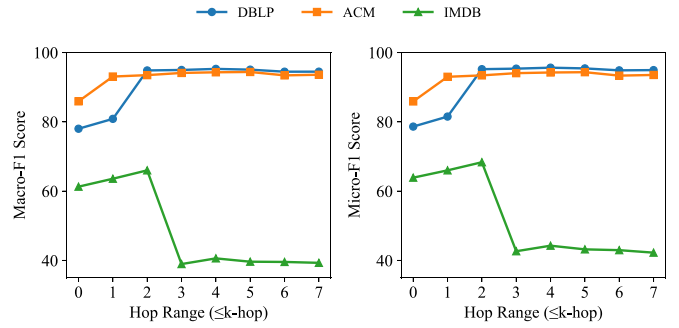


Fig. 4. Node Classification Performance with Varying Hop Ranges on DBLP, ACM, and IMDB.

mere type differentiation cannot capture fine-grained topological roles. HAN aggregates neighbors through predefined meta-paths but also underperforms, revealing it primarily encodes type-level connectivity semantics rather than structural roles. SeHGNN adopts a pre-computation paradigm, completely bypassing topological learning during message-passing, and also fails on Hetero-House, validating that its success relies on attribute discriminability rather than topological modeling. HGSL optimizes graph topology end-to-end but performs worst, as its structure learning relies on feature similarity, which may even destroy discriminative topological structures when features lack semantics. In contrast, MTHGN employs only simple adjacency aggregation yet achieves comparable performance to sophisticated models.

Combining results from Tables III and IV, we find that on attribute-rich real-world datasets, simple adjacency aggregation achieves superior performance, while on topology-sensitive Hetero-House, most models fail to demonstrate effective structural learning capabilities. These observations suggest that prevalent HGNN evaluation protocols struggle to differentiate between shallow adjacency exploitation and genuine structural pattern learning.

E. Impact of Varied-Hop Neighbors

In this experiment, we evaluate the performance of neighbor aggregation strategies with varying hop ranges on node classification. Specifically, **0-hop aggregation** considers only the features of the node itself, and **≤ k-hop aggregation** extends this to incorporate features from all neighbors within k hops. We set the experimental scope to **≤ 7-hops** to observe the complete evolutionary trajectory of performance. The results are illustrated in Fig. 4.

DBLP exhibits gradual improvement from **0-hop**, gains moderate improvement at **≤ 1-hop**, and achieves a dramatic leap at **≤ 2-hops** before reaching peak performance at moderate hop counts and subsequently plateauing. ACM shows rapid saturation where a single expansion from **0-hop** to **≤ 1-hop** accomplishes the majority of performance gains, reflecting strong local homophily. IMDB displays a strikingly different and alarming pattern. Performance steadily improves from 0-hop and reaches a local peak at **≤ 2-hops**. However, extending to **≤ 3-hops** triggers a catastrophic performance collapse, and performance remains depressed at subsequent hop counts.

In conclusion, all three datasets exhibit clear optimal hop ranges beyond which performance plateaus or deteriorates. If models genuinely learned complex structural patterns across multiple hops, we should expect sustained improvement rather than rapid saturation after a few hops. The fact that performance peaks within such limited hop ranges indicates that current benchmarks primarily evaluate local feature aggregation capability rather than deep topological reasoning ability.

TABLE III
NODE CLASSIFICATION RESULTS ON DBLP, ACM, IMDB AND HETERO-HOUSE DATASETS

	DBLP		ACM		IMDB		Hetero-House	
	Macro-f1	Micro-f1	Macro-f1	Micro-f1	Macro-f1	Micro-f1	Macro-f1	Micro-f1
RGCN	91.52±0.50	92.07±0.50	91.55±0.74	91.41±0.75	58.85±0.26	62.05±0.15	31.67±5.41	37.30±1.64
HAN	91.67±0.49	92.05±0.62	90.89±0.41	90.79±0.43	57.74±0.96	64.63±0.58	27.03±2.75	37.33±2.04
HGT	93.01±0.23	93.49±0.25	91.12±0.76	91.00±0.76	63.00±0.19	67.20±0.57	32.13±1.05	43.10±0.91
SimpleHGN	94.01±0.24	94.46±0.22	93.42±0.44	93.35±0.45	63.53±1.36	67.36±0.57	29.03±1.11	<u>39.57±0.86</u>
HGSL	81.82±0.55	81.23±0.63	88.09±0.65	88.23±0.64	39.31±0.58	35.61±0.67	18.63±0.59	38.80±1.71
SeHGNN	95.06±0.17	95.42±0.17	94.05±0.35	93.98±0.36	67.11±0.25	<u>69.17±0.43</u>	<u>32.00±1.00</u>	36.00±2.00
MTHGN	<u>94.87±0.18</u>	<u>95.23±0.17</u>	<u>93.74±0.54</u>	<u>93.66±0.55</u>	<u>66.98±0.68</u>	69.57±0.57	31.70±1.25	35.54±2.06

TABLE IV
HIGHER-ORDER RELATION PREDICTION RESULTS ON DBLP, ACM, AND IMDB DATASETS

	DBLP			ACM			IMDB		
	NMI	Macro-F1	Micro-F1	NMI	Macro-F1	Micro-F1	NMI	Macro-F1	Micro-F1
RGCN	56.68 ± 9.40	75.75 ± 7.93	76.47 ± 7.60	70.27 ± 5.20	75.12 ± 3.94	76.79 ± 3.31	85.97 ± 4.50	94.38 ± 2.54	94.41 ± 2.52
HAN	29.87 ± 3.35	40.48 ± 8.20	47.35 ± 3.72	40.85 ± 7.75	37.02 ± 8.12	47.70 ± 5.72	59.05 ± 2.83	52.81 ± 12.46	61.00 ± 7.87
HGT	33.16 ± 9.70	28.41 ± 4.74	41.82 ± 5.02	46.91 ± 1.29	35.01 ± 6.53	48.71 ± 4.10	64.74 ± 1.68	55.07 ± 13.83	63.06 ± 8.62
SimpleHGN	33.25 ± 7.96	49.22 ± 6.83	54.82 ± 6.29	51.24 ± 8.82	49.63 ± 12.49	56.79 ± 9.21	74.82 ± 6.29	80.05 ± 10.82	82.60 ± 8.08
HGSL	69.34 ± 3.65	85.49 ± 1.83	85.82 ± 1.89	76.13 ± 2.60	76.77 ± 2.14	78.61 ± 1.33	94.31 ± 1.04	98.13 ± 0.42	98.14 ± 0.42
SeHGNN	<u>78.22 ± 0.93</u>	<u>91.07 ± 0.55</u>	<u>91.19 ± 0.55</u>	<u>76.82 ± 0.95</u>	<u>82.47 ± 2.69</u>	<u>82.67 ± 2.45</u>	<u>95.17 ± 1.02</u>	98.72 ± 0.30	98.73 ± 0.30
MTHGN	79.63 ± 1.35	91.49 ± 0.63	91.66 ± 0.62	77.19 ± 0.54	84.01 ± 0.85	84.06 ± 0.85	95.55 ± 1.08	<u>98.69 ± 0.35</u>	<u>98.70 ± 0.35</u>

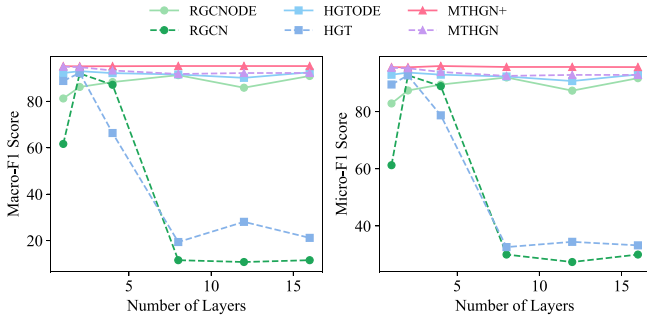


Fig. 5. Node Classification Performance of HGNNs and their ODE Version Under Different Numbers of Layers on the DBLP dataset.

F. Receptive Field Expansion Effect of HGNNs

To further test whether current experimental protocols can truly evaluate topological information, we design experiments that examine whether enlarging the topological receptive field of HGNNs improves performance. If the protocols are reliable, a larger receptive field should yield better results. However, the usual way to achieve this, namely stacking more graph convolution layers, leads to serious over-smoothing. This limitation prevents a fair test of the benefit of deeper topological information. To overcome this, we adopt an ODE-based formulation, which can enlarge the receptive field while maintaining model performance.

Fig. 5 illustrates the performance comparison of various models at different layers on the DBLP dataset. As the number of discrete layers increases, the Macro-F1 and Micro-F1 scores of traditional RGCN and HGT models show a notable decline. In contrast, RGCN (ODE) and HGT (ODE) demonstrate remarkable stability and consistently outperform their discrete counterparts across different depths. This performance improvement may be attributed to two possible reasons:

first, the ODE formulation effectively mitigates the over-smoothing problem inherent in stacked graph convolution layers; second, the inherent capability of ODEs to model complex dynamics may independently contribute to performance gains.

To determine which factor is responsible, we propose MTHGN+, an ODE version of MTHGN. Since MTHGN does not employ graph convolution layers, it is inherently free from over-smoothing issues. Therefore, comparing MTHGN+ with MTHGN allows us to evaluate the second possibility. As illustrated in Fig. 5, MTHGN+ does not substantially outperform MTHGN, indicating that the modeling capability of ODEs alone is not the primary reason for performance improvement. Consequently, the superior performance of RGCN (ODE) can be attributed to the ODE formulation's ability to mitigate over-smoothing, thereby enabling the expansion of the receptive field and the capture of richer topological information.

Furthermore, as the depth increases, while HGT (ODE) and RGCN (ODE) consistently maintain high performance, their performance does not significantly surpass the peak levels achieved by HGT and RGCN in their shallow layers. Although the ODE-based formulation enables the model to capture more extensive topological information, this enriched structural knowledge yields nearly negligible performance improvements on downstream tasks. Moreover, the combination of pre-computation and MLP is already capable of achieving optimal results, while overly intricate designs may instead lead to redundancy.

V. DISCUSSION

Our experiments reveal limitations in current HGNN evaluation protocols from multiple perspectives. We introduced MTHGN as a minimal topology baseline, yet it achieves near SOTA results on real-world datasets. Conversely, on the topology-sensitive Hetero-House dataset, HGNNs fail to demonstrate effective structural learning capabilities. These contrasts confirm that strong homophily in real-world datasets

makes shallow aggregation highly effective, masking models' lack of genuine topological understanding.

Based on these findings, we advocate redesigning evaluation protocols: 1) construct topology-sensitive benchmark datasets that systematically disentangle structural cues from attribute information, with our synthetic Hetero-House dataset serving as a preliminary attempt in this direction; 2) introduce topology-related tasks and metrics, such as higher-order relation prediction and structural evaluation metrics like NMI.

Our findings also explain the success of GNN-to-MLP knowledge distillation: on current benchmarks, teacher models primarily rely on attributes and local similarity, enabling MLPs to effectively inherit predictions; however, such distillation may fail on genuinely topology-dependent tasks.

VI. CONCLUSION

Most HGNNs claim to handle both attribute and topological information, yet this capability remains insufficiently verified. We conduct systematic experiments to examine this claim. First, we introduce MTHGN as a minimal topology baseline that uses only direct adjacency for feature aggregation. Surprisingly, it achieves near SOTA performance on real-world datasets. In contrast, all models perform poorly on the topology-sensitive Hetero-House dataset. Furthermore, we expand HGNNs' receptive fields through ODEs to capture deeper structural information, but this expansion does not yield significant performance improvements. These results demonstrate that current HGNN evaluation protocols struggle to effectively distinguish shallow adjacency exploitation from genuine structural pattern learning.

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REFERENCES

- [1] Y. Wu et al., "Graph convolutional networks with Markov random field reasoning for social spammer detection," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, pp. 1054–1061.
- [2] F. Qin et al., "CADGCL: Unsupervised retrieval of CAD models via boundary representations," *Vis. Comput.*, vol. 41, no. 9, pp. 6493–6505, 2025.
- [3] Z. Chen et al., "Can graph neural networks count substructures?," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 10383–10395.
- [4] Y. Xie, J. Jia, C. Wen, D. Li, and M. Li, "Multi-topology contrastive graph representation learning," *Sci. China Inf. Sci.*, vol. 69, no. 2, 2026, Art. no. 122102.
- [5] G. Yang, M. Li, H. Feng, and X. Zhuang, "Deeper insights into deep graph convolutional networks: Stability and generalization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 48, no. 2, pp. 1707–1719, Feb. 2026.
- [6] C. Gao et al., "A survey of graph neural networks for recommender systems: Challenges, methods, and directions," *ACM Trans. Recommender Syst.*, vol. 1, no. 1, pp. 1–51, 2023.
- [7] S. Kojaku et al., "Network community detection via neural embeddings," *Nature Commun.*, vol. 15, no. 1, 2024, Art. no. 9446.
- [8] C. Morris et al., "Weisfeiler and leman go neural: Higher-order graph neural networks," in *Proc. AAAI Conf. Artif. Intell.*, 2019, vol. 33, pp. 4602–4609.
- [9] G. Bouritsas et al., "Improving graph neural network expressivity via subgraph isomorphism counting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 657–668, Jan. 2023.
- [10] M. Zhang and P. Li, "Nested graph neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 15734–15747.
- [11] P. A. Papp et al., "DropGNN: Random dropouts increase the expressiveness of graph neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 21997–22009.
- [12] H. Liu, P. Jiao, X. Guo, H. Wu, M. Gao, and J. Zhang, "HGN2T: A simple but plug-and-play framework extending HGNNs on heterogeneous temporal graphs," *IEEE Trans. Big Data*, vol. 10, no. 5, pp. 620–632, Oct. 2024.
- [13] X. Wang, D. Bo, C. Shi, S. Fan, Y. Ye, and P. S. Yu, "A survey on heterogeneous graph embedding: Methods, techniques, applications and sources," *IEEE Trans. Big Data*, vol. 9, no. 2, pp. 415–436, Apr. 2023.
- [14] L. Xhonneux, M. Qu, and J. Tang, "Continuous graph neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 10432–10441.
- [15] Z. Hu et al., "Heterogeneous graph transformer," in *Proc. Web Conf.*, 2020, pp. 2704–2710.
- [16] Q. Lv et al., "Are we really making much progress? Revisiting, benchmarking and refining heterogeneous graph neural networks," in *Proc. 27th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2021, pp. 1150–1160.
- [17] P. Veličković et al., "Graph attention networks," in *Proc. Int. Conf. Learn. Representations*, 2018. [Online]. Available: <https://openreview.net/forum?id=rJXMPikCZ>
- [18] Y. Wang et al., "HeTa: Relation-wise heterogeneous graph foundation attack model," in *Proc. 34th Int. Joint Conf. Artif. Intell.*, 2025, pp. 3453–3461.
- [19] X. Fu et al., "MAGNN: Metapath aggregated graph neural network for heterogeneous graph embedding," in *Proc. Web Conf.*, 2020, pp. 2331–2341.
- [20] X. Fu and I. King, "MECCH: Metapath context convolution-based heterogeneous graph neural networks," *Neural Netw.*, vol. 170, pp. 266–75, 2024.
- [21] J. Fan et al., "Path-aware multi-scale learning for heterogeneous graph neural network," *Neural Netw.*, vol. 191, 2025, Art. no. 107743.
- [22] J. Zhao et al., "Heterogeneous graph structure learning for graph neural networks," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 5, pp. 4697–4705.
- [23] Q. Long et al., "HGK-GNN: Heterogeneous graph kernel based graph neural networks," in *Proc. 27th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2021, pp. 1129–1138.
- [24] X. Han, Z. Feng, and Y. Ning, "A topology-aware graph coarsening framework for continual graph learning," in *Proc. 38th Int. Conf. Neural Inf. Process. Syst.*, Red Hook, NY, USA: Curran Associates Inc., 2024, pp. 132491–132523.
- [25] X. Yang et al., "Simple and efficient heterogeneous graph neural network," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 9, pp. 10816–10824.
- [26] J. Hu, B. Hooi, and B. He, "Efficient heterogeneous graph learning via random projection," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 12, pp. 8093–8107, Dec. 2024.
- [27] Z. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: Generating explanations for graph neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, vol. 32, pp. 9240–9251.
- [28] X. Wang et al., "Heterogeneous graph attention network," in *Proc. World Wide Web Conf.*, 2019, pp. pp. 2022–2032.
- [29] M. Schlichtkrull et al., "Modeling relational data with graph convolutional networks," in *Proc. Eur. Semantic Web Conf.*, 2018, pp. 593–607.
- [30] X. Guo et al., "Counterfactual learning for higher-order relation prediction in heterogeneous information networks," *Neural Netw.*, vol. 183, 2025, Art. no. 107024.



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